

Assignment - 8

1. Basic Test Case:

Write a unittest class `TestMath` with one test method that checks if $2 + 3$ equals 5.

2. Setup and Teardown:

Create a test class that uses `setUp()` to initialize a list `[1, 2, 3]` and `tearDown()` to print "Test completed."

Verify that the list length is 3 inside your test method.

3. Multiple Assertions:

Write a test class `TestStringMethods` with methods to test: `"hello".upper()` equals "HELLO" `"hello".isupper()` returns False

4. Exception Testing:

Use `assertRaises` to verify that dividing by zero ($10 / 0$) raises a `ZeroDivisionError`.

5. Test Suite Execution:

Create two test classes (`TestAdd` and `TestSubtract`) and combine them into a single test suite using `unittest.TestSuite`. Run the suite using `unittest.TextTestRunner()`.

Solution:

I have written code in two different files that is `src→math_calculations.py` and `test→test_math_calculations.py`

`math_calculations.py`

```
def add(a, b):  
    return a + b  
  
def subtract(a, b):  
    return a - b  
  
def divide(a, b):  
    return a / b
```

`test_math_calculationsa.py`

```
import unittest  
from Homework.src.math_calculations import add, subtract, divide  
  
# 1. Basic Test Case
```

```

class TestMath(unittest.TestCase):
    def test_add_simple(self):
        self.assertEqual(add(2, 3), 5)

# 2. Setup and Teardown
class TestList(unittest.TestCase):
    def setUp(self):
        self.data = [1, 2, 3]

    def tearDown(self):
        print("Test completed")

    def test_list_length(self):
        self.assertEqual(len(self.data), 3)

# 3. Multiple Assertions
class TestStringMethods(unittest.TestCase):
    def test_upper(self):
        self.assertEqual("hello".upper(), "HELLO")

    def test_isupper(self):
        self.assertFalse("hello".isupper())

# 4. Exception Testing
class TestExceptions(unittest.TestCase):
    def test_zero_division(self):
        with self.assertRaises(ZeroDivisionError):
            divide(10, 0)

# 5. Test Suite Execution Classes
class TestAdd(unittest.TestCase):
    def test_run_add(self):
        self.assertEqual(add(10, 5), 15)

class TestSubtract(unittest.TestCase):
    def test_run_subtract(self):
        self.assertEqual(subtract(10, 5), 5)

if __name__ == "__main__":
    # Create the Test Suite
    suite = unittest.TestSuite()

    # Add specific classes to the suite

```

```

suite.addTest(unittest.makeSuite(TestAdd))
suite.addTest(unittest.makeSuite(TestSubtract))

# Run the suite
runner = unittest.TextTestRunner()
runner.run(suite)

```

Outputs:

"C:\Wipro Training\UnitTest\UnitTestTraining\venv\Scripts\python.exe" "C:/Program Files/JetBrains/PyCharm Community Edition 2025.2.6/plugins/python-ce/helpers/pycharm/_jb_unittest_runner.py" --path "C:\Wipro Training\UnitTest\UnitTestTraining\Homework\test\test_math_calculations.py"

Testing started at 2:29 AM ...

Launching unittests with arguments python -m unittest C:\Wipro Training\UnitTest\UnitTestTraining\Homework\test\test_math_calculations.py in C:\Wipro Training\UnitTest\UnitTestTraining

Test completed

